

Yurii Zaika

AI / Python Engineer

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Ten years of production engineering plus a personal track in Python and ML. Built a real-time speech pipeline: faster-whisper on CUDA, pyannote diarization, VAD and streaming, with regression measurements of latency and RTF. Alongside it: data pipelines on SQLAlchemy and Playwright, and daily work with AI agents.

AT A GLANCE

- 10 years — in production retail systems, 2016 to now
- 33 — internal systems of the network carry my code
- 27 — repositories in 2026: freelance and personal projects
- 6 sources — in the listings pipeline: OLX, LUN, Rieltor, Metrazh, Telegram, FB

WHAT I DELIVERED

- Built a real-time speech pipeline in Python: faster-whisper on GPU, pyannote diarization, VAD and streaming capture.
- Wrote its quality regression suite — RTF, latency, noise robustness, accuracy ceilings — and drive decisions from the numbers rather than impressions.
- Built Python data pipelines: SQLAlchemy 2 and Alembic over PostgreSQL, Playwright for protected pages, six listing sources merged into one catalog.

STACK

Data: MS SQL Server, T-SQL, PostgreSQL

Engineering practice: Legacy modernization, Production ownership

Python: Python 3.11, FastAPI, SQLAlchemy 2, Alembic, Pydantic, Typer CLI, Playwright, selectolax, NumPy, pytest, ruff / black, Jinja2

Speech and ML: faster-whisper (CUDA), pyannote.audio, VAD, Speaker diarization, Quality evaluation: RTF, latency, Streaming audio pipelines, WASAPI / sounddevice, PyTorch / GPU

Working with AI agents: Claude Code, MCP-сервери, Custom subagents and hooks, Agent tooling (CDP), Prompt engineering

JS / TS ecosystem: Next.js 16, React 19, NestJS, Prisma, Node.js

KEY PROJECTS

Real-time speech pipeline (ORK) (2026) — Author

Real-time speech recognition from calls: parallel microphone and system-audio capture, VAD-based segmentation, faster-whisper on GPU, pyannote diarization with incremental clustering, translation. A separate regression suite — RTF, latency under load, noise robustness — shows exactly where quality is lost.

Python 3.11, faster-whisper, pyannote.audio, CUDA, FastAPI, NumPy, WASAPI

Real-estate data pipeline (2026) — Author

A listings parser with pagination, normalization and price history: SQLAlchemy 2 models with Alembic migrations over PostgreSQL, two transports — plain HTTP and Playwright for protected pages — a Typer CLI, pydantic-settings config and pytest coverage.

Python, SQLAlchemy 2, Alembic, PostgreSQL, Playwright, selectolax, Typer, pytest

Legacy system map (2026) — Author

A tool for mapping the trading system before migration: Python parses .dfm and .pas files, extracts stored-procedure calls and screen relations, and a Next.js app renders it as an annotated map.

Python, Next.js, TypeScript, PostgreSQL

AM Mobility — web and mobile app (2026) — Full stack developer

Client project: a NestJS back end with Swagger and PostgreSQL, a Next.js web app, and an Expo / React Native mobile app with MapLibre maps, geolocation and offline state.

NestJS, Next.js, Expo, React Native, PostgreSQL, MapLibre, TanStack Query

EXPERIENCE

Freelance and personal projects — AI / Python and Full Stack Developer

2026-03 — present · Dnipro · remote

- A real-time speech pipeline: faster-whisper on CUDA, pyannote diarization, VAD, WASAPI capture.
- And a regression suite for it: RTF, latency under load, noise robustness, quality ceilings — decisions come from measurements, not impressions.
- A real-estate parser in Python: SQLAlchemy 2 and Alembic over PostgreSQL, Playwright and selectolax, a Typer CLI, pytest and ruff.
- A listings pipeline over six sources with geocoding, deduplication and catalog publishing through GitHub Actions.
- A legacy-code analyzer: Python parses .dfm and .pas files and diffs stored-procedure calls, Next.js renders the system map with annotations.
- Client work: NestJS with Prisma and PostgreSQL, Next.js 16 with React 19, a mobile app on Expo with MapLibre maps.
- Daily work with AI agents: MCP servers, custom subagents and hooks, and my own CDP browser inspector as agent tooling.

ATB-Market — Lead Application Programmer

2016-06 — present · Dnipro

- Since 2018 also build internal services on ASP.NET Core and Angular: access, audit, reporting, integrations — 33 of the network's systems carry my code.
- Network integrations: EDI and e-document flow, M.E.Doc, Oracle EBS, 1C, Kafka, BizTalk, SSIS and SSRS.
- Engineering practice: xUnit and NSubstitute tests, SonarQube and Jenkins, Serilog, Docker, a private NuGet feed.

EDUCATION AND CERTIFICATES

- Computer Engineering · computer systems and networks — Zaporizhzhia Institute of Economics and Information Technologies (2008 — 2011)
- Mining Engineer, underground development — Kryvyi Rih Technical University (1998 — 2003)
- Angular 9 — Luxoft (2021)

CODE ON GITHUB

- Housing catalog across nine regions — six-source aggregator with a map, price indices and a GitHub Actions build
<https://github.com/qwert11/budynky-zahid>
- ML Formula Wiki — 72 machine-learning formulas from Raschka's book, rendered with KaTeX
<https://github.com/qwert11/ML-Formula-Wiki>
- M3 to M4 Mapping Checker — diffs UI and stored-procedure calls between Delphi 6 and Delphi 13
https://github.com/qwert11/d6_d13_dbmapping_checker
- TeamBeacon — a real-time beacon for finding your group in a crowd: ASP.NET Core, SignalR, Docker
<https://github.com/qwert11/TeamBeacon>
- Resume Engine — this site: YAML to Jinja to HTML, PDF, DOCX and Markdown in two languages
<https://github.com/qwert11/resume2>

LANGUAGES

- Ukrainian — native
- English — technical documentation and written communication; spoken at basic level